# Glycan-mediated interactions of infectious hypodermal and hematopoietic necrosis virus (IHHNV) with host tissues in Pacific whiteleg shrimp (*Penaeus vannamei)*

**Key words**: arthropoda/ligands/O-glycan/virus

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**Running Title**: Glycan ligands of IHHNV in Pacific whiteleg shrimp

**Supplementary data**: Supplementary Figure 1; Supplementary Table 1

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**Abstract**

Infectious hypodermal and hematopoietic necrosis virus (IHHNV) poses a significant threat to global shrimp aquaculture, causing substantial economic losses due to Runt Deformity Syndrome (RDS) in susceptible species, particularly *Penaeus vannamei*. Despite its widespread impact, the molecular mechanisms underlying IHHNV's tissue tropism and host entry remain poorly understood. This study investigated the role of host glycans in mediating IHHNV attachment and infection, focusing on the interaction between virus-like particles (VLPs) of IHHNV and glycan profiles in *P. vannamei* tissues. Using recombinant IHHN-VLPs, we demonstrated preferential binding to muscle and gill tissues, with significantly higher affinity compared to the hepatopancreas. Lectin-based assays and mass spectrometry revealed that these tissues were enriched in oligomannosylated *N*-glycans substituted or not with fucose, terminal N-acetylglucosamine (GlcNAc) and sulfate groups as well as complex *O*-glycans showing very unusual structural features such as substituted fucose residues and terminal *N*-acetylgalactosamine residues. Inhibition assays on ELISA, Western blot and histochemistry on shrimp's tissues modified with lectins (ConA, WGA, AAL) and glycosidases (PNGase-F, mannosidase, fucosidase) demonstrated that identified mannosylated and fucosylated glycans were essential for IHHNV recognition. These findings highlight the importance of glycan-mediated interactions in IHHNV infection, suggesting that high-mannose *N*-glycans and fucosylated *O*-glycans serve as primary attachment factors. The conservation of these glycan motifs across crustacean species implies their potential as broad-spectrum viral targets. This study not only advances our understanding of IHHNV pathogenesis but also opens avenues for developing glycan-based antiviral strategies, such as competitive inhibitors or targeted glycosidase treatments, to mitigate viral outbreaks in shrimp aquaculture.

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## Introduction

Viral diseases have caused substantial economic losses to the global shrimp-farming industry (Vega-Heredia et al. 2012). More than 20 viruses are known to infect cultured and wild shrimps (Yu et al. 2021), among which infectious hypodermal and haematopoietic necrosis virus (IHHNV) is one of the most serious threats to aquaculture and fisheries (Vega-Heredia et al. 2012). First reported in cultured *Penaeus stylirostris* (Pacific blue shrimp) in Hawaii in 1981(Lightner et al. 1983), IHHNV has since spread worldwide and is now reported in more than 20 countries, including Thailand (Kibenge et al. 2024). IHHNV belongs to the Parvoviridae family and carries a single-stranded DNA genome enclosed in a non-enveloped, icosahedral capsid. It is the causative agent of Runt deformity syndrome (RDS), a condition characterized by stunted growth, cuticular deformities, and high mortality in juvenile shrimp (Lightner et al. 1983). RDS can cause up to 50% production losses in *Penaeus* aquaculture, and the global economic impact of IHHNV has led to its inclusion on the list of notifiable crustacean pathogens by the World Organization for Animal Health (OIE) (Lightner et al. 2012). This virus infects several penaeid species, including *P. vannamei* (Pacific whiteleg shrimp), *P. stylirostris*, *P. monodon* (giant tiger prawn), and *P. japonicus* (kuruma shrimp), as well as non-penaeid crustaceans such as *Macrobrachium rosenbergii* (Bangladeshi freshwater prawn), crayfish, crabs, and even bivalve shellfish, with epidemiological patterns and disease severity varying across hosts (Bell and Lightner 1984; Bonami et al. 1990; Rai et al. 2012; Yu et al. 2021). Post-larval and juvenile shrimps are particularly vulnerable, showing infection rates nearly twice those of adults. Among the susceptible hosts, we focused on *P. vannamei* because it is the most widely cultivated shrimp species worldwide and thus the most relevant model for studying the impact of the disease on aquaculture (Bell and Lightner 1984; Kalagayan et al. 1991; Rai et al. 2009; Anderson et al. 2017).

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At the tissue level, IHHNV exhibits strong tropism toward ectodermal- and mesodermal-derived tissues, such as gills, cuticular epithelium, hematopoietic, and lymphoid organs, while sparing endodermal tissues including the hepatopancreas, midgut, and smooth muscle (Lightner 2011). These viruses thrive in environments with rapid host growth, as they preferentially target actively dividing cells. This dependency arises because parvoviruses lack their own DNA polymerase and rely entirely on the host's replication machinery (Jager et al. 2021). Moreover, differences in susceptibility among host developmental stages and tissues indicate that host-specific factors, such as surface receptors and innate immune responses, play a critical role in shaping IHHNV infectivity and tissue tropism.

IHHNV shares structural similarities with other parvoviruses, featuring a T=1 icosahedral symmetry and a capsid composed of multiple viral protein (VP) subunits (Bonami et al. 1990; Mari et al. 1993). The capsid proteins are critical for host cell attachment and internalization, yet the specific host receptors mediating IHHNV entry remain poorly characterized. Given that IHHNV infects multiple tissues but causes the most severe pathology in developing cuticular and hematopoietic cells, it is plausible that the virus may sequentially orchestrate its interaction with variable proteinaceous or glycosylated host receptors for modulating its entry. This hypothesis is partly supported by studies of diverse non-enveloped viruses that utilize glycan-mediated binding for internalization into the host cells. For instance, noroviruses (*Caliciviridae*) and their virus-like particles (VLPs) demonstrate specific glycan-dependent interactions, particularly with histological blood group antigens (HBGAs) and sulfated glycolipids for cellular entry (Huang et al. 2005; Tsukamoto et al. 2024). Similarly, *Macrobrachium rosenbergii* nodavirus (MrNV) has been shown to bind specifically to fucosylated *N*-glycans in the gills of *M. Rosenbergii* (Somrit et al. 2020). Glycan mediated interaction has also been recognized across various virus families: dsRNA

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*Reoviridae* utilize either HBGAs or neuraminic acid (NeuAc) terminal glycans to a broad range of hosts, ranging from mammals to insects, while ssDNA *Parvoviridae* and *Papovaviridae* depend on heparan sulfate proteoglycans for cell entry (Huang et al. 2014). Despite these advances in understanding viral entry mechanisms, the identification of the host surface receptor(s) involved in IHHNV attachment and entry, as well as the subsequent molecular mechanisms, has not yet been reported. A preliminary clue within the parvovirus family suggests that *N*-linked glycans and glycosaminoglycans (GAGs) may play a crucial role to modulate virus-host interaction (Huang et al. 2014; Stencel-Baerenwald et al. 2014; Raman et al. 2016; Shanker et al. 2017; Thompson et al. 2019; Bally et al. 2021) . These insights highlight the evolutionary conservation of glycans as viral attachment factors across diverse host species. By shaping viral entry and host specificity, glycan–virus interactions provide a critical framework for understanding viral tropism. Extending this perspective to shrimp offers an opportunity to uncover how similar mechanisms operate in aquatic hosts and how they may influence susceptibility to infection.

## RESULTS

### IHHN-VLP exhibited binding affinity in shrimp muscles and gills

IHHN-VLP used in this study was designed by inserting a full sequence of IHHNV capsid protein (accession# AF218166.2) into the PET17b plasmid that encoded 330 amino acids (Fig. 1A) with an expected molecular mass of 32 kDa. The IHHN-VLP capsid protein was obtained with a purity greater than 95% and a yield of 2 mg of purified protein per liter of *E. coli* lysate (Fig. 1A). The recombinant IHHNV capsid protein was successfully biotinylated, as confirmed by Western blotting with streptavidin (Fig. 1B), prior to its use in multiple binding assays. Importantly, the biotinylated IHHNV capsid protein retained its ability to self-assemble into icosahedral virus-like particles (VLPs) of approximately 27 nm in diameter (Fig. 1C), consistent with the architecture reported for the parental IHHN-VLP (Kiatmetha et al. 2018).



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To evaluate IHHN-VLP binding on shrimp tissues, whole-mount assays were performed on *P. vannamei* post-larvae and revealed distinct tissue tropism. Control shrimp displayed only minimal background fluorescence (Fig. 2A), whereas those incubated with IHHN-VLP exhibited strong fluorescent signals, predominantly in the muscle and gill tissues. These results indicate that IHHN-VLP specifically targets these organs *in situ*.

***Quantitative and qualitative IHHN-VLP binding to tissue lysates***

An ELISA was performed to quantify IHHN-VLP interactions with four tissue lysates from *P. vannamei*: hepatopancreas, muscle, stomach and gill tissues (Fig. 2B). Binding was significantly higher in muscle ($0.662 \pm 0.043$) and gill ($0.785 \pm 0.057$) tissues compared to the hepatopancreas ($0.293 \pm 0.036$) ($P \leq 0.001$). In addition, gill tissue exhibited a higher mean binding level than muscle ($P \leq 0.05$). These quantitative ELISA results corroborate the whole-mount observations, confirming that muscle and gill tissues are primary targets of IHHN-VLP binding (Fig. 2A).

To further characterize these interactions, Far-Western blot analysis was conducted on the four tissue extracts. IHHN-VLP bound to multiple protein bands with molecular masses of 150, 100, 75, 40, 35 and 30 (Fig. 2C, left panel), suggesting that several host proteins may act synergistically or in concert with the invading virions. In contrast, control blots in which tissues were incubated with streptavidin alone showed only minimal background signal (Fig. 2C, right panel).

***Host tissues contain protein-bound fucosylated- and high-mannose glycans***

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To determine whether IHHNV utilizes glycans as its binding ligands, we first characterized the glycan profiles of four tissue extracts in *P. vannamei* using a panel of lectins with defined specificities (Fig. 3).

First, quantitative analysis with protein extracts from gills (Gil), muscles (Mus), hepatopancreas (Hep) and stomach (Sto) was performed by ELISA using a set of six lectins (Fig. 3A). *Sambucus nigra* lectin (SNA-I), which is known to specifically recognize to sialic acid attached to terminal galactose in α-2,6 and to a lesser degree, α-2,3 linkage, was used as a negative control based on the knowledge that sialic acids are largely absent in Arthropoda (Tiemeyer et al. 2015). As expected, SNA-I showed a very low binding to the four tissue extracts, serving as base signal intensity. In contrast, wheat germ agglutinin (WGA) known to bind *N*-acetylglucosamine and *N*-acetylgalactosamine, Concanavalin A lectin (ConA) known to bind glucose and mannose, and *Aleuria aurentia* lectin (AAL), known to bind fucose residues, showed robust bindings to both gill and muscle tissues (Bojar et al. 2022). Out of these, ConA lectin exhibited the strongest overall labelling signal towards protein extracts from gills and muscles, with O.D values of 1.411±0.021 and 0.882±0.046, respectively. Lower yet robust binding intensities ranging from 1.154±0.064 to 1.684±0.110 were observed for AAL and WGA lectins in the gill extracts, and from 0.599±0.017 to 0.916±0.113 in the muscle extracts. *Ricinus communis* agglutinin (RCA), known to bind β-galactosides, such as LacNAc, or *N*-acetylgalactosamine residues showed significant binding in gills (0.740±0.034) but very weak in muscles (0.313±0.012). Finally, *Ulex europaeus* lectin (UEA-I) known to preferentially bind Fucα-1,2 residues such as in human blood group antigens, did not show any binding (0.224±0.054 in gills; 0.214±0.003 in muscles) in an equivalent manner to SNA-I (0.179±0.016 in gills; 0.194±0.007 in muscles). These findings collectively strongly support the presence of complex glycans harboring terminal mannose, GlcNAc/GalNAc fucose residues,

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although with a notable absence of Fuc-α-1,2 linkages and sialic acid, in the muscle and gill tissues of *P. vannamei*.

ELISA data were then confirmed by Western blot analyses using the highest binders, WGA, ConA and AAL. As shown in Figure 3B, WGA and ConA showed intense binding with sets of well-defined protein bands from both tissues, ranging from 37 kDa and more than 100 kDa. In contrast, AAL is strongly bound to a set of proteins around 50-55 kDa in gills but showed a diffuse and weaker recognition pattern for high molecular weight components in muscles.

Altogether, ELISA and Western blot analyses demonstrate the presence of complex fucosylated and mannosylated glycans substituting numerous glycoproteins in the two tissues that may serve as main targets for IHHNV binding. To confirm the results of the lectin analysis, we aimed to identify the glycans associated with proteins expressed in gill and muscle tissues, particularly the *N*- and *O*-linked glycans usually involved in virus binding. To achieve this, total glycoproteins were extracted from the tissues and digested by protease, followed by the sequential release of *N*- and *O*-glycans from the glycopeptides. The *N*-glycans, released in a reducing form by peptide:N-glycosidase F (PNGase F), were labelled with the fluorescent probe procainamide prior to analysis by LC-FL-MS/MS. Analysis by fluorescence HPLC coupled to mass spectrometry of the *N*-glycans from both tissues revealed multiple intense signals (Fig. 4). The retention times and monosaccharide compositions deduced from the *m/z* values allowed to identify 15 major *N*-glycans (Table 1). Seven of these were identified as oligomannose type *N*-glycans, composed of two HexNAc residues and three to nine Hex residues. MS/MS fragmentation confirmed the presence of a $HexNAc_2$ disaccharide associated with the procainamide group, which was further substituted by three to nine Hex residues. Furthermore, two intense signals at retention times (RT) 30.43 and 31.90 min with *m/z* values of 2265 and 2427, respectively, consistent with



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$HexNAc_2Hex_{10}$ and $HexNAc_2Hex_{11}$ compositions, were observed in muscle. Based on their *m/z* values and their RT compared with standards from mammalian cell lines, these two signals were associated with the commonly observed $GlcNAc_2Man_9(Glc)_{1-2}$ *N*-glycans, which are typically regarded as biosynthetic intermediates. These two signals could be observed in the gills, but only in very small quantities. Three signals were also identified as fucosylated glycans ($GlcNAc_2Man_2Fuc_1$, $GlcNAc_2Man_3Fuc_1$ and $GlcNAc_3Man_3Fuc_1$) based on their respective *m/z* values of 1114, 1276 and 1479 at RT 8.63, 12.07 and 14.15 min. MS/MS analysis of all three *N*-glycans revealed that the fucose residue was substituting for the reducing GlcNAc residue, as evidenced by the presence of a prominent $[M+H]^+$ fragment ion at *m/z* 587 ($HexNAc_1Fuc_1$) and a minor fragment ion at *m/z* 790 ($HexNAc_2Fuc_1$) (Fig. 5). Additionally, the presence of a distinctive M-HexNAc fragment at *m/z* 1276 in the MS/MS pattern of $GlcNAc_3Man_3Fuc_1$ was indicative of a GlcNAc residue in the terminal non-reducing position. Finally, a signal at RT of 10.19 minutes showed an *m/z* value of 1356, which corresponded to a sulphated version of the $GlcNAc_2Man_3Fuc_1$ *N*-glycan. MS/MS analysis of this compound revealed an identical fragmentation pattern to $GlcNAc_2Man_3Fuc_1$, with a prominent M–80 fragment at *m/z* 1276, indicating the loss of the $SO_3$ group. The absence of any sulphated fragments in the MS/MS pattern following this type of fragmentation suggests that the $SO_3$ group is located on one of the two terminal Man residues.

Following the liberation of *N*-glycans by PNGaseF, reductive β-elimination was used to release the *O*-glycans from the remaining glycopeptides. This produced reduced glycans that were not amenable to labelling with procainamide. The reduced *O*-glycans were thus permethylated prior to analysis by MALDI-MS and MALDI-MS/MS mass spectrometry, as previously described. (Yamakawa et al. 2018). As shown in Fig. 6A, three major fucosylated reduced *O*-glycan signals at *m/z* 882.3, 953.4 and *m/z* 1127.5, annotated as $Fuc_2Hex_1HexNAc_1$, $Fuc_1Hex_1HexNAc_2$ and

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$Fuc_2Hex_1HexNAc_2$, were identified in gills. By contrast, we could not identify *O*-glycan signals in muscle with high confidence, which is likely due to their lower quantity. This is consistent with the fact that, like all epithelia, gills are covered by layers of mucin-type glycoproteins rich in *O*-glycans. MS/MS analysis of $Fuc_2Hex_1HexNAc_1$-ol signal at *m/z* 882 showed two intense signals at *m/z* 502 and 520 which, when coupled with the loss of 218 mass units (m.u.) at *m/z* 664, demonstrated the presence of a Galβ1,6GalNAc-ol *O*-glycan core structure. Furthermore, the presence of the C/Z fragment ions couple at *m/z* 403/502 was indicative of the presence of a very unusual fucose dimer. This epitope was shown to be linked to the C3 position of the GalNAc-ol, as indicated by the very intense z fragment ion at *m/z* 502. This oxonium-type fragment ion allows unambiguous differentiation of the C-3 position of GalNAc and GalNac-ol from other positions (Yamakawa et al. 2018; Auger et al. 2023). However, the fragment ion at *m/z* 694, indicative of the release of terminal Fuc residue, cannot inform us on its linkage to the external Fuc residue at this stage as the oxonium fragment ion cannot be produced from the Fuc-Fuc linkage, irrespective of its position. Altogether, these data allowed the identification of a very unusual difucosylated *O*-glycan with the most probable sequence being Fucα1,xFucα1,3(Galβ1,6)GalNAc-ol, where x could not be unambiguously established from the MS/MS data. The MS/MS fragmentation spectrum of the *O*-glycan $Fuc_1Hex_1HexNAc_2$-ol at *m/z* 953 was characterized by the fragment ions at *m/z* 735 and 694 that demonstrate the presence of both the Galβ1,6GalNAc-ol core and a terminal HexNAc residues. Then, the very intense C/Z fragment ions couple at *m/z* 474/502 established the presence of the HexNAc-Fucα1,3 motif substituting the GalNAc-ol residue. The *O*-glycan at *m/z* 953 was thus identified as a variation of the previous one in which the terminal Fuc residue was exchanged by a HexNAc residue to form the HexNAc1,xFucα1,3(Galβ1,6)GalNAc-ol sequence. Finally, the third major component at *m/z*

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1127 was identified a combination of the two previous ones in which the difucosylated motif observed in the glycan at *m/z* 882 was further substituted by a terminal HexNAc, as observed in the glycan at *m/z* 953. This hypothesis was supported by the observation of a very intense C fragment ion at m/z 648 indicative of a $HexNAcFuc_2$ motif that can further lose a terminal HexNAc residue, as shown by the secondary fragment ion at *m/z* 389. To further characterize *O*-glycans, particularly the linkage pattern of fucose residues and identification of terminal monosaccharides, the permethylated *O*-glycan fraction was subjected to linkage analysis as partially methylated alditol acetate (PMAA) derivatives by gas chromatography coupled with mass spectrometry (GC-MS). The nature of the individual PMAAs was determined based on their fragmentation patterns in electronic impact (EI) mode and their RT, compared to authentic standards where available. (York et al. 1986). All fragmentation patterns were also compared with those in the open access database of PMAAs from the Complex Carbohydrate Research Center at the University of Georgia (https://ccrc.uga.edu/). Of particular interest, two fucose derivatives were identified at RT of 16.4 and 20.4 minutes, respectively: a fucose in the terminal non-reducing position (t-Fuc) and a fucose substituted in the C2 position (2-Fuc) (Fig. 6C). The t-Fuc residue was identified by comparing its RT to that of an authentic standard, whereas the 2-Fuc residue was identified exclusively by its EI-MS spectrum due to the absence of an authentic standard. No trace of fucose substituted in the C-3 or C-4 positions could be established based on prediction of their EI-MS spectra. This confirmed that internal fucose residues were exclusively substituted in the C-2 position. Finally, the presence of terminal galactose (t-Gal) and terminal GalNAc (t-GalNAc) was confirmed by their EI-MS spectra and RT (Fig. 6C). It should be noted that, for reasons that remain to be elucidated, we were unable to directly identify the reduced 3,6-linked GalNAc PMAA residue in this analysis. However, the observation of intense C/Y fragment ions for all three compounds at m/z s403/502,

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474/502 and 648/502 clearly demonstrated that the fucosylated branch was linked through C3 linkage to the GalNAc residue. In accordance with the observations made, the exclusive observation of Y-type fragment for Hex branch has demonstrated that the t-Gal is linked through a C6 linkage. Identification of the 2-Fuc, t-Fuc, t-Gal and t-GalNAc residues, together with MS/MS analysis, enabled further characterization of the three identified *O*-glycans as the most probable sequences: Fucα1,2Fucα1,3(Galβ1,6)GalNAc-ol, GalNAc1,2Fucα1,3(Galβ1,6)GalNAc-ol, and GalNAc1,2Fucα1,2Fucα1,3(Galβ1,6)GalNAc-ol. It should be noted that we could not determine the anomeric configuration of the GalNAc residue that substitutes for two of these glycans in the terminal non-reducing position.

Overall, the *N*- and *O*-glycan profiles established by LC-FLR-MS, MALDI-MS and GC-MS are in line with the lectin profiling that indicated the presence of terminal fucose, GlcNAc and/or GalNAc and mannose residues. It is important to note that although we have documented by GC-MS and MALDI-MS that terminal αFuc1,2 epitope was present on *O*-glycans, it was not detected by UEA-I. This strongly suggests that the recognition of Fucα1-2Gal epitope by UEA-I does not extend to the Fucα1-2Fuc unusual epitope (Bojar et al. 2022).

***IHHN-VLP binding is dependent on glycans***

Having confirmed that the binding affinity of IHHN-VLP varied significantly among the tissues, we further dissected the specificity of the glycan recognition of IHHN-VLP, by a lectin inhibition assay using ELISA assay and histochemistry. As shown in Fig 7A, binding of IHHN-VLP to muscle tissues was significantly inhibited by WGA, ConA, AAL as well as RCA ($P \leq 0.0001$) to a lesser extent, whereas UEA-I and SNA-I showed no significant inhibition in



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accordance with the glycan composition of gills. In muscles, WGA, AAL, RCA and ConA significantly inhibited IHHN-VLP binding ($P \leq 0.0001$), while UEA-I, and SNA-I showed no significant inhibition (Fig. 7A). These results strongly suggest that the binding of IHHN-VLP to *P. vannamei* tissues is mediated by orchestral glycan interactions, with *N*-acetylglucosamine/*N*-acetylgalactosamine, mannose, and fucose residues being the prominent glycans in an individual organ. Such glycosylation may also influence IHHNV infection tropism or selectivity in shrimp tissues.

To further emphasize the importance of specific glycans as targets for IHHN-VLP, the sugar moieties in tissue extracts were chemically and enzymatically digested, and followed ELISA assays with IHHN-VLP. In this regard, gill and muscle tissue lysates were pre-treated either with sodium periodate ($NaIO_4$, a general oxidant that cleaves vicinal diols in carbohydrates), PNGase-F, exo-mannosidase, or exo-fucosidase enzymes prior to its incubation with IHHN-VLP. The muscle lysate pre-treated with $NaIO_4$ resulted in the extinction of VLP binding ($P \leq 0.0001$) (Fig. 7B), confirming the need of carbohydrate for IHHN-VLP binding. Treatments of PNGase-F, mannosidase and fucosidase over gill tissue extract revealed a similar trend of significant reduction in IHHN-VLP binding ($P \leq 0.0001$), albeit to a lesser extent than $NaIO_4$. The binding of IHHN-VLP to these three enzyme-treated extracts ranged from about 40% to 60% compared to untreated. Similarly in gill tissue, the treatments with the three enzymes resulted in a significant reduction of IHHN-VLP binding, consistent with those found in muscle tissue (Fig. 7B).

The involvement of glycans in IHHNV binding was confirmed in gill and muscle tissues using histochemistry. Consistent with ELISA and Western blot data (Fig. 3 and 7A), ConA, WGA, RCA and AAL lectins were found to strongly bind to the gill epithelium (Fig. 7C). Furthermore, pre-incubation of these four biotinylated lectins significantly inhibited the binding of IHHN-VLPs



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compared to tissues that were not pre-incubated. A similar observation was made on muscle tissue, although lectin binding and the inhibition of IHHN-VLP binding by lectin pre-incubation were lower, except for WGA lectin (Fig. 7D).

Finally, to try to gain deeper insights into the carbohydrate-binding specificity, we tested IHHNV-VLP binding on the Consortium for Functional Glycomics' (CFG's) glycan micro array containing 585 different carbohydrate epitopes. It should be noted that the commercial glycan micro array is exclusively constituted of mammalian derived compounds, that may not be totally relevant when glycans ligands expressed in invertebrates like those observed in *Penaeus vannamei* tissues. Binding of IHHNV-VLP was detected for a very narrow range of three bi-antennary *N*-glycans elongated by at least two LAcNac motifs (Fig. 8). that only contained terminal type 2 LacNAc motifs, while type 1 were notrecognized. Along with these three oligosaccharides, a single fucosylated epitope terminated by a pseudo type 2 LacNAc Galβ1,4GalNAc was identified as a possible ligand.

**Discussion**

The present report provides new insight into the molecular interactions of IHHNV in *P. vannamei*. By using IHHN-VLPs, which display the virus's exposed capsid proteins, to probe glycan-mediated virus-host interactions, we demonstrated that IHHNV preferentially binds to muscle and gill tissues through the glycan epitopes that are predominantly expressed on these host tissues (Figs 2). Glycan profiling using lectin-based assays and mass spectrometry showed that both gill and muscle tissues are enriched in mannose (which is Con-A lectin preference), fucose (AAL) and terminal GlcNAc/GalNAc (WGA) containing glycoproteins (Fig. 3). Along with this line of evidence, the significant inhibition of IHHN-VLP binding by ConA, AAL, and WGA



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lectins and by specific glycosidases confirmed the functional importance of these glycans in virus recognition. In addition, MALDI-MS results further confirmed that *N*-glycans in gill tissue were dominated by oligomannosylated species and core-fucosylated structures. The selective binding pattern may correspond with previous reports showing that IHHNV preferentially infects ectodermal- and mesodermal-derived tissues, such as gills, cuticular epithelium, and hematopoietic organs, rather than endodermal-derived tissues like the hepatopancreas.

Selective glycan-binding patterns are also reported for other crustacean viruses. For instance, *Macrobrachium rosenbergii* nodavirus (MrNV) was shown to preferentially bind to fucosylated *N*-and *O*-glycans in the gill tissue of its host (Somrit et al., 2020). Beyond MrNV, glycan-dependent binding has also been described for white spot syndrome virus (WSSV). One major envelope protein, VP24, directly binds chitin (poly-GlcNAc) in the shrimp digestive tract; mutational mapping (aa 186–200) identified the chitin-binding region, and a peptide mimicking this site plus oligo-chitosan significantly blocked WSSV–chitin interactions and reduced per os infection *in vivo* (Li et al. 2016). Separately, the envelope protein VP37 shows high-affinity binding to heparin-like sulfated glycosaminoglycans on shrimp hemocytes, an interaction inhibited by sulfated galactan; crystallography of the VP37 C-terminal domain (2.51 Å) revealed discrete sulfate binding sites (R213, K257) required for heparin engagement by SPR (Sotanon et al. 2018; Somsoros et al. 2021). Together with the MrNV fucosylated glycans example, these findings strengthen the view that carbohydrate motifs, chitin/GlcNAc, fucosylated *N*-glycans, and sulfated GAGs, serve as conserved attachment factors in crustacean virology, providing more insights to explain tissue tropism at exposed epithelia (gut, gill, cuticle) where such glycans are abundant (Sotanon et al. 2018; Somrit et al. 2020; Somsoros et al. 2021). The shared carbohydrate motifs between *P. vannamei* and *M. rosenbergii* suggest that fucosylated glycans glycans may represent

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a conserved mechanism for viral interactions in crustaceans, serving as common attachment factors across species. However, these glycan motifs are not unique to shrimp; complex *N*-glycans containing mannose and fucose are also abundant in vertebrates, including humans, yet IHHNV is not infectious to humans (Lightner 2011).

To further characterize glycosylation in *P. vannamei*, *O*-glycans were analyzed by mass spectrometry and GC/MS analyses. The analysis revealed three major fucosylated *O*-glycans (*m/z* 882.3, 953.4, 1127.5), all containing a Galβ1,6GalNAc-ol core substituted with internal fucose residues substituted in C-2 position and, in some variants, terminal GalNac residue. The *m/z* 953 and 1127 species further highlight structural diversity arising from replacement or extension of terminal residues while maintaining the conserved difucosylated core. These results are consistent with the lectin-binding data, in which AAL recognized fucosylated and RCA binds GalNAc/GlcNAc-terminated structures in gill tissues. Interestingly, identical MS signals at *m/z* 953 and 1127 were observed from our previous study in the *O*-glycans fraction isolated from the gills of the marine shrimp *Macrobrachium rosenbergii*. MS/MS analysis demonstrated that these two *O*-glycans were substituted by a terminal H-type1 trisaccharide (Fuc$\alpha$1,2Galβ1,3GlcNAc) and by a Lewis b tetrasaccharide (Fuc$\alpha$1,2Galβ1,3(Fuc$\alpha$1,4)GlcNAc), respectively. It was thus tempting to associate the two signals in *P. vannamei* to these glycans previously identified. However, MS/MS of the permethylated *O*-glycans newly isolated from *P. vannamei* surprisingly revealed that those compounds were isomeric structures of the previously identified glycans with very unusual sequences characterized by the presence of internal fucose residues and fucose dimers. Similar motifs have also been observed in other crustaceans, such as *Artemia franciscana*, where glycosphingolipids contain fucosylated LacdiNAc with α-GlcNAc termini and branched fucose (Kojima et al. 2011). Beyond crustaceans, substituted fucose residues have also been

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identified in *N*-glycans of other species, where internal fucose residues themselves are substituted by GlcA and GlcN residues in the blue mussel *Mytilus edulis* (Zhou et al. 2013) or as antennal modifications in the marine snail *Volvarina rubella* (Eckmair et al. 2016). These examples underscore that fucose branching is not an isolated feature but can be found across diverse glycomes. Supporting the functional relevance of fucosylation in shrimp, recent studies have shown that *P. vannamei* hemocyanin (PvHMC) undergoes fucosylation mediated by fucosyltransferases (PvFUT8/10), which enhances its antibacterial activity and binding to Gram-negative bacteria, whereas defucosylation reduces its activity (Jiang et al. 2025). The possibility that IHHNV engages both glycans and proteinaceous receptors is consistent with mechanisms described for other parvoviruses. In many cases, glycans serve as low-affinity, high-avidity attachment factors that concentrate virions on the cell surface, enabling subsequent engagement of specific protein receptors that trigger entry and uncoating (Stencel-Baerenwald et al. 2014). The far-Western blot results here, which revealed VLP binding to multiple protein bands of varying molecular weights, raise the possibility that IHHNV recognizes a diverse set of glycoprotein complexes. Identifying these proteins will be an important step toward fully elucidating the entry pathway.

Glycan array experiment revealed a small number of potential ligands almost exclusively constituted by polymerized type-2 LacNAc motifs, which is well in agreement with the observed impact of RCA lectin, known to interact with these motifs, on virus attachment to tissues and proteins. However, the absence of mannosylated and fucosylated ligands recognized on the array despite their presence in shrimp tissues and the significant effect of AAL and CoNA on the virus binding strongly suggests that IHHNV is highly specific to motifs absent from the micro array, which is mainly constituted by mammalian-inspired glycans. This is particularly the case for

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fucosylated *O*-glycans which exhibited highly unusual motives consisting of fucose dimers such as Fucα1,2Fuc, or of fucose residues substituted by GalNAc such as GalNAc1,2Fuc which are not represented on any glycan microarray. The absence of mannosylated glycans in the potential targets on the glycan array is more difficult to explain at the present time given the presence of consensual oligomannosylated *N*-glycans in muscles and gills, suggesting that the virus might demonstrate a specificity for a subfamily of mannosylated glycans absent from the microarray among those. In this respect, a recent study have reported that a range of innate immune receptors known to bind to oligomannosylated *N*-glycans exhibit, in fact, unique, and sometimes very restricted, specificities toward oligomannose motifs (Gao et al. 2021). To address this issue, we are right currently developing a natural glycan array strategy using a bifunctional fluorescent linker that is conjugated through its arylamine group by reductive amination, in order to determine the specificity of IHHNV towards its *in vivo* substrates (Song et al. 2009). This approach has recently been successfully applied to determine the exact specificity of anti-carbohydrate antibodies employing *N*-glycans and free glycans (Nuschy and Wilson 2025). However, it is highly challenging in the present case because the glycans can only be purified in minute amount from natural sources. Furthermore, building a microarray from *O*-glycans is also technically challenging due to the difficulty of efficiently releasing *O*-linked glycans in a reducing form, which explains why such a strategy has, to the best of our knowledge, never been applied to mucin-type *O*-glycans up to know. An alternative approach would be to chemically synthetize the highly unusual GalNAc1,2Fucα1,3GalNAc, and GalNAc1,2Fucα1,2Fucα1,3GalNAc epitopes, in order to directly assay their interaction with IHHNV VLP.

From a disease management perspective, understanding that IHHNV targets high-mannose and fucosylated glycans has opened up the venues for intervention. Glycan-mimetic inhibitors,



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competitive lectin analogs, or targeted glycosidase treatments might be developed to interfere with viral attachment in aquaculture systems. Similar strategies have been tested in other viral systems, including noroviruses and human parvoviruses, where blocking the glycan-binding step can substantially reduce infectivity (Huang et al. 2014; Thompson et al. 2019). Several lectins show how carbohydrate-binding proteins can block viruses by targeting high-mannose or sulfated glycans on the viral surface. The red-algal lectin griffithsin (GRFT) and the cyanobacterial lectin cyanovirin-N (CV-N) both bind clusters of high-mannose *N*-glycans and can neutralize viruses at low-nanomolar doses by preventing attachment and fusion. Other reported lectins with similar activity include KAA-2, HRL40, and plant lectins such as GNA and UDA (Akkouh et al. 2015; Alvarez et al. 2023). While these lectins were developed mainly for human/animal viruses, the same glycan-blocking concept can be adapted to shrimp if safety is confirmed. Research has demonstrated the value of targeting glycans. For instance, sulfated galactans from *Gracilaria fisheri* (not lectins but polyanionic glycan decoys) have been shown to enhance resistance to WSSV *in vivo*. This resistance is mechanistically aligned with VP37–heparin-like binding, as previously described. (Rudtanatip et al. 2014; Somsoros et al. 2021). In this context, screening exogenous lectins for shrimp safety, spectrum against IHHN-VLP binding, and delivery via microencapsulation represents a rational path to glycan-focused prophylaxis in hatcheries (Wongprasert et al. 2014; Akkouh et al. 2015; Fernández Romero et al. 2021; Alvarez et al. 2023). However, such approaches would need to be carefully designed to avoid negative effects on host physiology, given the essential roles of glycans in normal cell function.

Conclusively, this study demonstrates that IHHN-VLP binding to *P. vannamei* tissues is strongly influenced by glycan composition, contributing to enhanced gill binding. In addition, mass spectrometry revealed three predominant fucosylated *O*-glycans, all based on a

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Galβ1,6GalNAc-ol core with branched fucose substitutions, which may further modulate viral recognition. While these glycans are critical for attachment, they are unlikely to be the sole determinants of host specificity, and protein receptors may also be required. The conservation of these glycan motifs across crustacean species suggests that they could act as common viral attachment factors, and their identification provides a valuable target for future antiviral strategies in shrimp aquaculture.

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## MATERIALS AND METHODS

### *Chemicals and reagents*

All biotinylated lectins including concanavalin A (ConA), succinylated wheat germ agglutinin (WGA), *Ricinus communis* agglutinin I (RCA-I), *Ulex europaeus* agglutinin I (UEA-I), and *Aleuria aurantia* lectin (AAL), SNA-I, *Sambucus nigra* Agglutinin I were from Vector Labs. Related antibodies and biotinylated streptavidin were purchased from Thermo Fisher Scientifics. Glyco-enzymes such as PNGase F, α1-2,3,6 mannosidase, and α1-2,3,4,6 fucosidase were procured from New England Biolabs while other general enzymes and monosaccharides used for inhibition studies were from Sigma-Aldrich. Other chemicals for Western blotting and recombinant protein production and other related buffers were mostly from EMD Ltd.

### *Preparation of IHHN-VLP*

Recombinant IHHN-VLP capsid protein was expressed and purified according to our previous publication (Weerachatyanukul et al. 2022). Briefly, pET16b plasmids (Novagen, Darmstadt, Germany) containing an inserted IHHNV capsid protein gene with a hexahistidine tag sequence (IHHNVCAP-6xHis) were transformed into competent *E. coli* BL21 (DE3). The bacterial culture was then inoculated into Lysogeny broth (LB) broth containing 50 μg/mL of ampicillin. Protein expression was induced with 1 mM IPTG (25°C, 3 h), and the bacterial cells were subsequently lysed in phosphate-buffered saline (PBS) containing a protease inhibitor cocktail (GE Healthcare, Piscataway, NJ). Bacterial cells were pelleted (6000 ×g, 5 min, 4°C) and then lysed using 1% sarkosyl in PBS followed by reciprocal sonication at 20 Hz and 100 W for 5 min. After centrifugation (12,000 ×g, 5 min, 4°C), the supernatant was filtered by 0.45 μm filter



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papers (Sigma) and applied onto a nickel nitrilotriacetic acid (NTA) His-trap chromatographic column for affinity purification. The IHHN-VLPs fractions were eluted using an elution buffer (20 mM Tris–HCl, 250 mM imidazole, 500 mM NaCl, pH 7.4) and subsequently desalted with a G25 desalting column (GE Healthcare, Piscataway, NJ) using PBS as running buffer.

***Validation of IHHN-VLP***

Crude *E. coli* extracts and purified IHHN-VLP fractions were resolved by 12% SDS-PAGE under denaturing conditions (Laemmli 1970). Parallel gels were prepared, with one stained by Coomassie Brilliant Blue (Invitrogen, Carlsbad, CA, USA) for protein visualization, while the other was reserved for Western blot analysis. For western blot analysis, gel was transferred to nitrocellulose membranes (wet transfer, 100–120 V, 1 h, 4°C), and blocked with 3–5% bovine serum albumin (BSA) in Tris Buffered Saline with Tween-20 (TBST) (1 h, room temperature). Membranes were probed with monoclonal anti-mouse antibody against Hexahistidine (6xHis) (1:5,000; overnight, 4°C), washed three times with TBST, incubated with horseradish peroxidase (HRP) conjugated anti-mouse antibody (1:10,000; 30–60 min, room temperature), and developed using an enhanced chemiluminescence or ECL kit (Pico). Identical exposure settings were maintained for lane-to-lane comparisons. Lastly, for TEM analysis, purified IHHN-VLPs (0.1–0.5 mg/mL in TBS, pH 7.8) were applied to carbon-coated copper grids, negatively stained with 2% uranyl acetate, and visualized using a JEOL 1230 transmission electron microscope (TEM) at 25,000× and 50,000× magnifications operated at 120 kV (JEOL, Tokyo, Japan). The images were acquired and processed by Gatan digital micrograph program.

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***Whole-mount bindings of IHHN-VLP in P. vannamei tissues***

The binding of IHHN-VLP to *P. vannamei* tissues was investigated using a whole-mount immunofluorescence. *P. vannamei* juveniles (approximately 5-7 gm each) obtained from a private shrimp farm in Ayutthaya province, Thailand. Shrimp were fixed overnight in 4% paraformaldehyde, followed by a 3-day decalcification at room temperature in 0.1 M hydrochloric acid/formic acid, and extensive washing with PBS. Biotinylated IHHN-VLP was prepared by mixing 1 mg of IHHN-VLP in 500 µL PBS (pH 7.4) with 1 mM NHS-X-biotin at room temperature for 1 hr. with agitation. Decalcified shrimp were then incubated for 2 days with 20 µg/mL biotinylated IHHN-VLP, followed by a 2-day incubation at 4°C with Alexa 488-conjugated streptavidin, before visualization under an Olympus AX123 epifluorescence microscope (Olympus, Tokyo, Japan).

***Immunohistochemistry for lectin binding and lectin inhibition***

Muscle and gill cryo-sections were used as a model for immunohistochemistry detection of lectin binding, and lectin inhibition of IHHN-VLP binding. Tissues were rehydrated in PBS (pH 7.4, 10 min) and circumscribed with a hydrophobic barrier. Sections were blocked with 1% horse serum in PBS (30 min, room temperature). For lectin binding, Tissues were incubated with biotinylated lectins, including ConA, AAL, WGA, and RCA-I (1:500) at 4 °C overnight, Alexa-488-conjugated streptavidin and counterstained with DAPI staining for nucleus identification. Whereas, for lectin-inhibition of IHHN-VLP binding, tissues were first pre-incubated with lectins, including ConA, AAL, WGA, and RCA-I at 4 °C overnight. Tissues then undergo extensive wash with PBS and incubated with IHHN-VLP (100 µg) at 4 °C overnight, followed by anti- mouse antibody against Hexahistidine (1:500) in incubation buffer (1% BSA, 1% goat serum, 0.3% Triton



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X-100, 0.01% sodium azide in PBS) 37 °C for 4 hours. After 3 × 15 min PBS washes, sections were labeled with anti-mouse secondary conjugated with Alexa 594 (1:1,000) for 30–60 min at room temperature (light-protected), washed 3 × 15 min in PBS, counterstained with DAPI. The sections were visualized under an Olypus FV1000 confocal laser scanning microscope using argon laser line with the emission filters of 522±5 nm (for Alexa 488) and 450±10 nm (for DAPI) (Olympus, Tokyo, Japan).

***Qualitative and quantitative binding assessment in tissue lysates***

Binding of IHHN-VLP in shrimp tissue lysates was assessed by Far Western blot and ELISA. Adult *P. vannamei* (approximately 60-65 gm each) were obtained from a private shrimp farm in Ayutthaya province, Thailand. Shrimp acclimatization and rearing were performed according to animal handling guidelines approved by the Animal Care Committee (IACUC), Mahidol University (Animal Ethics No. MUSC68-037-807). After collections of gill, hepatopancreas, stomach and muscle tissues, they were homogenized in RIPA buffer (50 mM Tris HCl, 150 mM NaCl, 1.0% (v/v) IGEPAL® CA-630, 0.5% (w/v) sodium deoxycholate, 1.0 mM ethylene-diethyl tetraacetic acid (EDTA), 0.1% sodium dodecyl sulfate (SDS), and 0.01% sodium azide) containing 0.5% Triton-X 100, ×1 protease-inhibitor cocktail (pH 7.4). For Far Western and lectin blots, 10 µg of tissue lysates were resolved by 10% SDS-PAGE and transferred to polyvinylidene fluoride (PVDF) membranes (Millipore, Billerica, MA). Membranes were blocked with 5% skim milk in PBS to prevent non-specific binding (room temperature, 2 h) followed by exposure to 10 µg/mL IHHN-VLP, 1:2,000 anti-His, prior to its exposure to 1:5,000 goat anti-mouse immunoglobulin G (IgG) secondary antibody conjugated with HRP (Abcam, Cambridge, UK). For lectin blot, the membrane was incubated with 4 µg/mL biotinylated lectins including



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WGA, ConA, AAL, RCA, UEA-I, and SNA-I (overnight, 4°C) followed by an exposure to streptavidin-HRP (1:2,500). Visualization of binding complex was performed by an Immobilon Crescendo Western HRP substrate (Amersham Biosciences, Piscataway, NJ).

For ELISA, approximately 100 µg/mL of each tissue protein lysate was diluted in coating buffer (0.1 M $Na_2CO_3$, 0.1 M $NaHCO_3$, pH 9.5) and coated onto 96-well microtiter plates (Maxisorp, ThermoFisher Scientific, Waltham, MA) overnight at 4°C. Coated proteins were incubated with 5% BSA in PBS to block non-specific binding and washed extensively with PBS containing 0.1% tween-20. They were further incubated with IHHN-VLP (overnight, 4°C) followed by an exposure to anti-Histag antibody (1:1,000) and then goat IgG conjugated with HRP. Alternatively for lectin blot, coated proteins were exposed to 4 µg/mL biotinylated lectins followed by streptavidin-HRP (1:2500). Visualization was achieved by incubating each well with SureBlue TMB peroxidase substrate (SeraCare, Milford, MA), and the reaction was terminated by addition of 1N sulfuric acid (1:1 ratio TMB substrate:sulfuric acid). Iintensity of the developed enzymatic products was detected at 452 nm wavelength using a VersaMax ELISA reader (Molecular devices, San Jose, CA).

***Lectin inhibition assay and enzymatic removal of glycans***

Inhibition assays were performed by fluorescent staining and in ELISA. The extracted proteins (on both membranes or 96-well plates (Maxisorp, ThermoFisher) were pre-incubated with 4 µg/mL lectins (overnight, 4°C) before the addition of 10 µg/mL IHHN-VLPs. Detection of IHHN-VLP binding with anti-His-tag antibody was performed according to the conditions described above.

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To evaluate the impact of glycan removal on IHHN-VLP binding, tissue lysates were subjected to chemical hydrolysis and enzymatic digestion with sodium periodate ($NaIO_4$, a general oxidant that cleaves vicinal diols in carbohydrates), PNGase-F (enzyme that hydrolyzes *N*-linked glycans), α1-2,3,6 mannosidase, or α1-2,3,4,6 fucosidase enzymes at 37°C overnight before being coated onto the 96-well plates. Thereafter, the coated proteins were processed likewise as for the ELISA assay described above to detect the binding of IHHN-VLP.

***Fractionation of N- and O-glycans and analysis by mass spectrometry***

*N*- and *O*-glycans were sequentially released as previously described (Somrit et al. 2020). In brief, tissues including gill and muscles were collected from shrimps, homogenized and then lyophilized. The lyophilized tissues were resuspended in a solution of 6 M guanidine hydrochloride and 5 mM EDTA in 0.1 M Tris/HCl, pH 8.6 and agitated at 4°C overnight. The extracted glycoproteins were reduced by addition of DTT at a final concentration of 10 mM for 1 hour at 37°C and then alkylated by addition of IAM at a final concentration of 50 mM at 37C for 1 hour in the dark. The alkylated proteins were desalted by using dialysis membrane (MWCO 6–8,000) followed by lyophilization. The recovered proteins were sequentially digested by trypsin in 50 mM ammonium bicarbonate buffer, pH 8.6 for 5 h before chymotrypsin digestion at 37°C overnight. After brief boiling and cooling down, the glycopeptides were subjected to PNGase F digestion at 37 C, overnight. *N*-glycans were fractionated from peptides mixtures by C18 Sep-Pak cartridge in 5% acetic acid. The retained peptides were eluted by 20–40% 1-propanol with 5% acetic acid, followed by drying. The eluted peptides were subjected to reductive elimination with 1 M sodium borohydride in 0.05 N NaOH at 37°C for 72 h. The reaction was stopped by acetic

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acid and sample was passed through Dowex 50W-X8. After drying, borate was distilled by co-evaporation of 10% acetic acid in methanol several times under a stream of nitrogen gas.

For LC-FLR-MS/MS analysis of *N*-glycans, the purified *N*-glycan samples were dried with a SpeedVac and labelled with procainamide, as previously described (Klapoetke et al. 2010). In brief, *N*-glycans were labeled with 400 mM procainamide (prepared in 3:7 ratio of acetic acid:DMSO (v/v) and 1 M sodium cyanoborohydride). The reaction mixture was heated at 65◦C for 3 h and the *N*-glycans purified on HILIC-SPE conditioned in 90% ACN water. To remove the excess labeling reagents, the sample was reconstituted in 90% ACN and loaded on HILIC-SPE, washed with 1mL 90% CAN and eluted with 300 µL 20% ACN. Procainamide-labelled *N*-glycans were analysed by LC-MS/MS on a Synapt-XS mass spectrometer (Waters™) equipped with a fluorescence detector. *N*-glycans were separated on a Acquity UPLC I-Class PLUS XBridge fitted with a Glycan BEH Amide column (2.5 µm, 2.1 mm×150 mm, Waters™) with a gradient of acetonitrile (solvent A) and 50mM ammonium formate (solvent B) as follows: flow rate 0,4 mL/min; 0,0 min, 75:25 (A:B); 40,0 min, 60:40; 42.5 min, 20:80; 47.5 min, 20:80, 50 min, 75:25; column temperature, 60°C; injection volume 10 µL; FLR Detector: λex: 310 nm λem: 370 nm. MS and MS/MS spectra were recorded in positive mode.

For MALDI analysis of *O*-glycans, the purified *O*-glycans were aliquoted for permethylation by using NaOH/DMSO method (Ciucanu and Kerek 1984). The per-methylated derivatives were then extracted in chloroform and repeatedly washed with water. Samples were prepared by mixing directly on the target 1 µL of oligosaccharide solution (1–5 pmol) with 1 µL of 2,5 dihydroxybenzoic acid matrix solution (10 mg/mL in $CH_3OH/H_2O$, 50/50, vol/vol). MALDI-TOF and MALDI-TOF/TOF spectra were acquired on a 4800 Proteomics Analyzer

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(Applied Biosystems, Framingham, MA, USA) and UltrafleXtreme (Bruker, Bremen) mass spectrometers in reflection positive mode.

***Linkage analysis as partially methylated alditol acetate derivatives***

The permethylated *O*-glycans were hydrolyzed with 4 M trifluoroacetic acid at 100 °C for 4 h, followed by reduction with 10 mg/mL sodium borodeuteride (NaBD4) for 16 h at room temperature. Subsequently, the resulting products were acetylated to yield the partially methylated alditol acetates (PMAA). Gas chromatography (GC) was performed on a Trace GC Ultra system (Thermo Scientific) equipped with a SOLGEL-1MS capillary column (30 m × 0.25 mm ID, 0.25 μm film thickness) and detected by a TSQ Quantum GC detector (Thermo Scientific). The carrier gas was helium (99.9995%) at a constant pressure of 100 kPa and an oven temperature of 120 °C. The temperature gradient was set from 120 °C to 230 °C over 55 min at a rate of 2 °C/min, and then increased to 270 °C in 4 min at a rate of 10 °C/min. An electron ionization source mode at 70 eV was used for the mass spectrometer operation. The mass spectral data and retention times of the PMAA derivatives of 4-GlcNAc, 3-GlcNAc, and non-reducing terminal GlcNAc were compared with those of authentic standard compounds for identification.

***Glycan microarray***

Glycan-binding specificity of IHHNV VLPs was evaluated using the National Center for Functional Glycomics (NCFG) printed glycan microarray platform (version 5.6), comprising 539 glycans printed in six replicates per glycan (https://research.bidmc.org/ncfg/service-collaboration-request) (Heimburg-Molinaro et al. 2024). VLPs were tested at 50 μg/mL and detected with an



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anti-His primary antibody followed by a fluorescent secondary antibody. Arrays were scanned by fluorescence, and signal intensities were processed using the MAWE workflow, which excludes the highest and lowest replicate values before calculating the mean relative fluorescence units (RFU) standard deviation, and coefficient of variation for each glycan (Mehta et al. 2024). The results have analyzed with GLAD (Glycan Array Dashboard), which enables comparison of binding profiles and identification of enriched glycan motifs (Mehta and Cummings 2019).

***Statistical Analysis***

All statistical analyses were performed in Python (version 3.10.12) using libraries including NumPy (version 2.0.2), Pandas (version 2.2.2), and SciPy (version 1.11.0). To evaluate differences among treatment groups relative to the control, a one-way Analysis of Variance (ANOVA) was applied. When the ANOVA indicated significance ($P \leq 0.05$), post-hoc comparisons were carried out using Dunnett's Multiple Comparison Test (via Scipy.Stats.dunnett). For pairwise comparisons between treatment groups and the control, Student's t-tests were additionally employed. A significance threshold of $P \leq 0.05$ was used throughout. Figures were generated with Matplotlib (version 3.7.1) and Seaborn (version 0.13.2).

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**Abbreviations**

AAL, *Aleuria aurantia* lectin; AIA, *Artocarpus integrifolia* agglutinin/Jacalin; ANOVA, Analysis of Variance; BSA, Bovine Serum Albumin; ConA, Concanavalin A; CV-N, Cyanovirin-N; DAPI, 4',6-Diamidino-2-phenylindole; DTT, Dithiothreitol; EDTA, Ethylenediaminetetraacetic acid; EI, Electronic Impact; ELISA, Enzyme-Linked Immunosorbent Assay; E. coli, *Escherichia coli*; Fuc, Fucose; Gal, Galactose; GalNAc, *N*-Acetylgalactosamine; GC-MS, Gas Chromatography-Mass



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Spectrometry; GlcNAc, *N*-Acetylglucosamine; GRFT, Griffithsin; HBGAs, Histological Blood Group Antigens; Hepes, 4-(2-Hydroxyethyl)-1-piperazineethanesulfonic acid; HPLC, High-Pressure Liquid Chromatography; HRP, Horseradish Peroxidase; IAM, Iodoacetamide; IHHNV, Infectious Hypodermal and Hematopoietic Necrosis Virus; IHHN-VLP, IHHNV Virus-Like Particle; IPTG, Isopropyl β-D-1-thiogalactopyranoside; LB, Lysogeny broth; LCA, *Lens culinaris* agglutinin; LC-MS/MS, Liquid Chromatography-Tandem Mass Spectrometry; MALDI-TOF, Matrix-Assisted Laser Desorption/Ionization-Time of Flight; MrNV, *Macrobrachium rosenbergii* nodavirus; MS, Mass Spectrometry; MS/MS, Tandem Mass Spectrometry; $NaIO_4$, Sodium Periodate; NeuAc, *N*-Acetylneuraminic Acid (Sialic Acid); NHS-X-biotin, *N*-Hydroxysuccinimide-biotin ; Ni-NTA, Nickel-Nitrilotriacetic Acid; OIE, World Organisation for Animal Health; P. vannamei, *Penaeus vannamei* (Pacific whiteleg shrimp); P. stylirostris, *Penaeus stylirostris* (Pacific blue shrimp); P. monodon, *Penaeus monodon* (Giant tiger prawn); P. japonicus, *Penaeus japonicus* (Kuruma shrimp); PBS, Phosphate-Buffered Saline; PMAA, Partially Methylated Alditol Acetate; PNGase F, Peptide-*N*-Glycosidase F; PNA, Peanut Agglutinin; PVDF, Polyvinylidene Fluoride; PvHMC, *Penaeus vannamei* Hemocyanin; RCA-I, *Ricinus communis* Agglutinin I; RDS, Runt Deformity Syndrome; SDS-PAGE, Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis; SNA-I, *Sambucus nigra* Agglutinin I; SPR, Surface Plasmon Resonance; TBST, Tris-Buffered Saline with Tween-20; TEM, Transmission Electron Microscopy; UEA-I, *Ulex europaeus* Agglutinin I; VP, Viral Protein ; WGA, Wheat Germ Agglutinin ; WSSV, White Spot Syndrome Virus.

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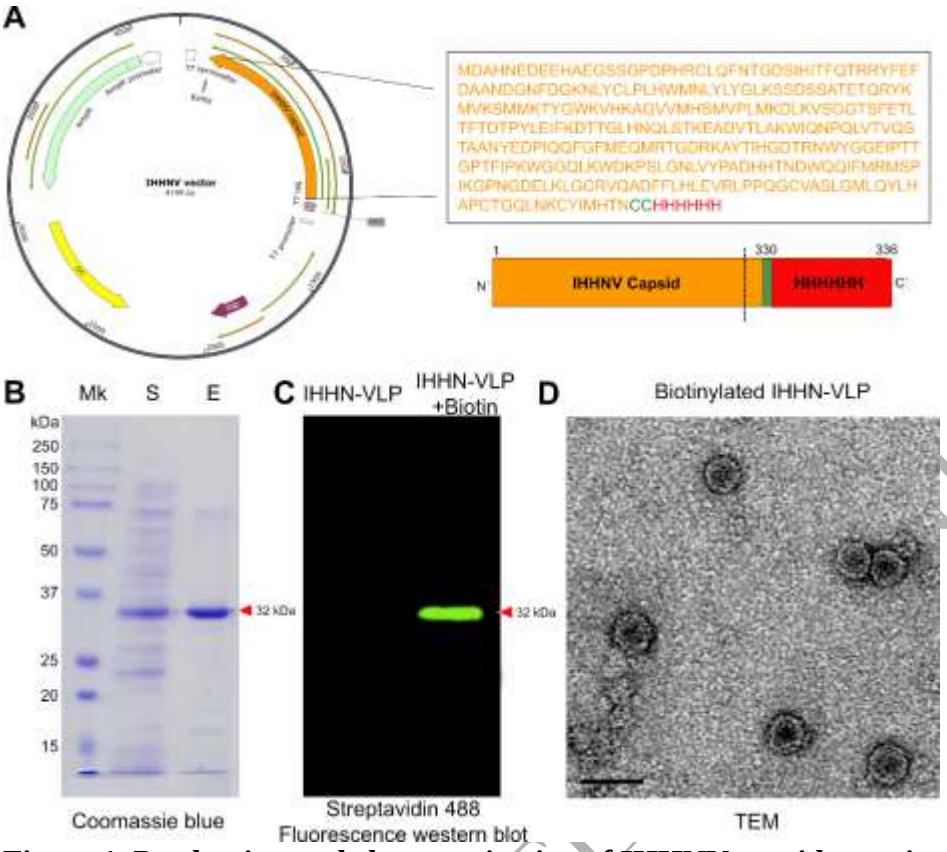



**Figure 1. Production and characterization of IHHNV capsid protein and VLPs**

(A) IHHN-VLP vector construct using pET17B plasmid as a template (left), which is inserted with the full sequence of IHHNV capsid gene (right). (B) SDS–PAGE of recombinant IHHNV capsid protein purified from *E. coli* BL21 (Rosetta). (C) Streptavidin–Alexa 488 Western blot confirming biotinylation of the purified capsid. (D) Transmission electron micrograph showing self-assembled IHHN-VLPs with ~27 nm icosahedral morphology. Scale bar is 30 nm.

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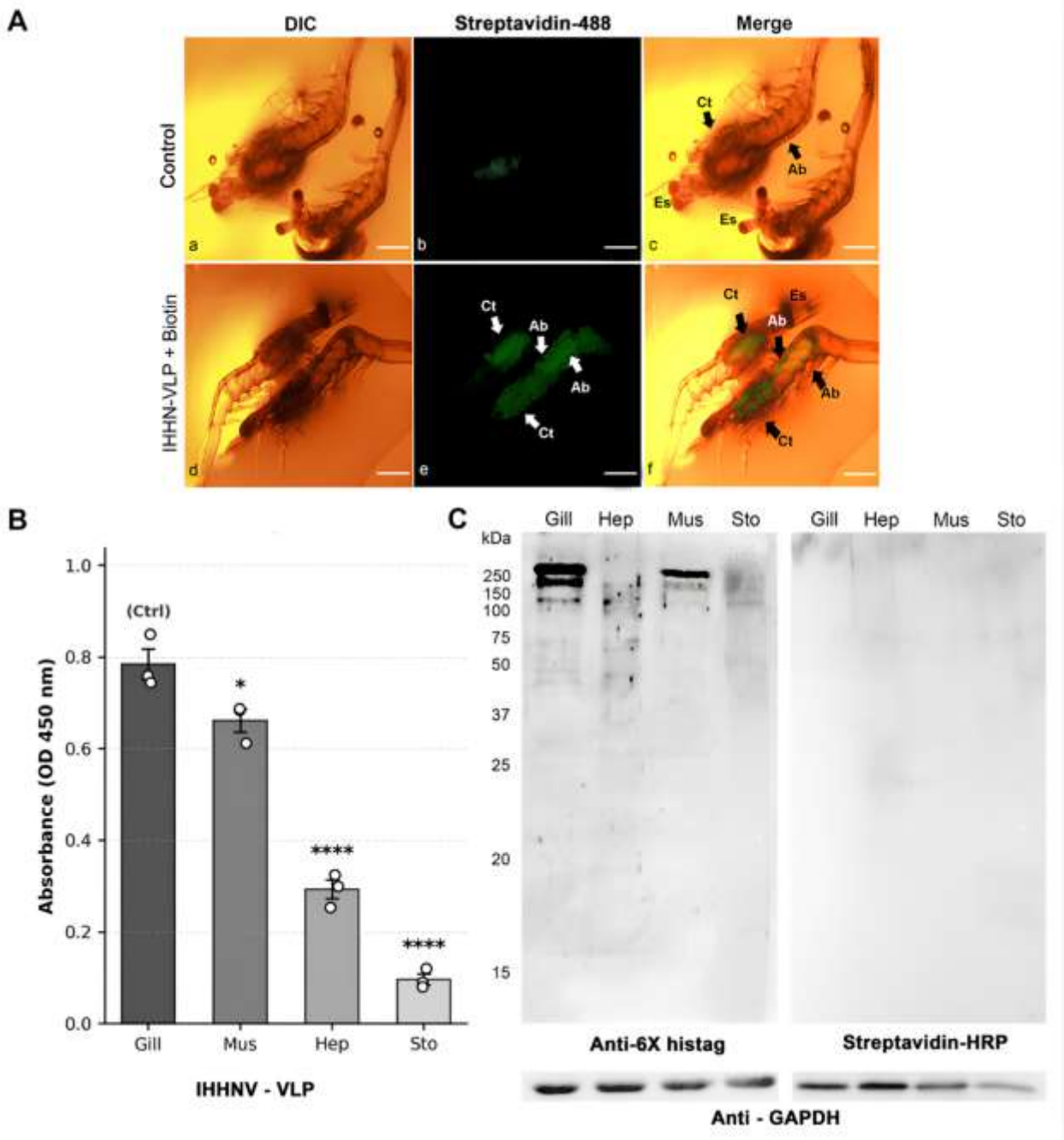



**Figure 2. IHHN-VLP binding to shrimp tissues in situ and ex vivo**

(A) Whole-mount fluorescence of *P. vannamei* post-larvae (control), showing minimal background (a, b, and c). Whole-mount fluorescence after incubation with IHHN-VLPs, showing strong signals localized to muscle and gill tissues, indicating tissue-specific targeting *in situ* (e and f) (Ct, cephalothorax; Es, eye stalk; Ab, abdomen). Scale bar = 1mm. (B) Quantification of IHHN-



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VLP binding to gill, muscle (Mus), hepatopancreas (Hep), and stomach (Sto) tissue lysates was carried out using ELISA. The bar graphs are presented as mean ±S.D. calculated from triplicate assays. Statistical comparisons were analyzed by one-way ANOVA followed by Dunnett's post-hoc test using gill as the reference control group (Ctrl). Significance levels are as follows: ****, $P \leq 0.0001$***, $P \leq 0.001$; *, $P \leq 0.05$. (C) Far-Western blot analysis of IHHN-VLP binding to shrimp tissue extracts on PVDF membrane which were then detected by anti-6X histag antibody (left panel) or streptavidin-HRP only as control (right panel).

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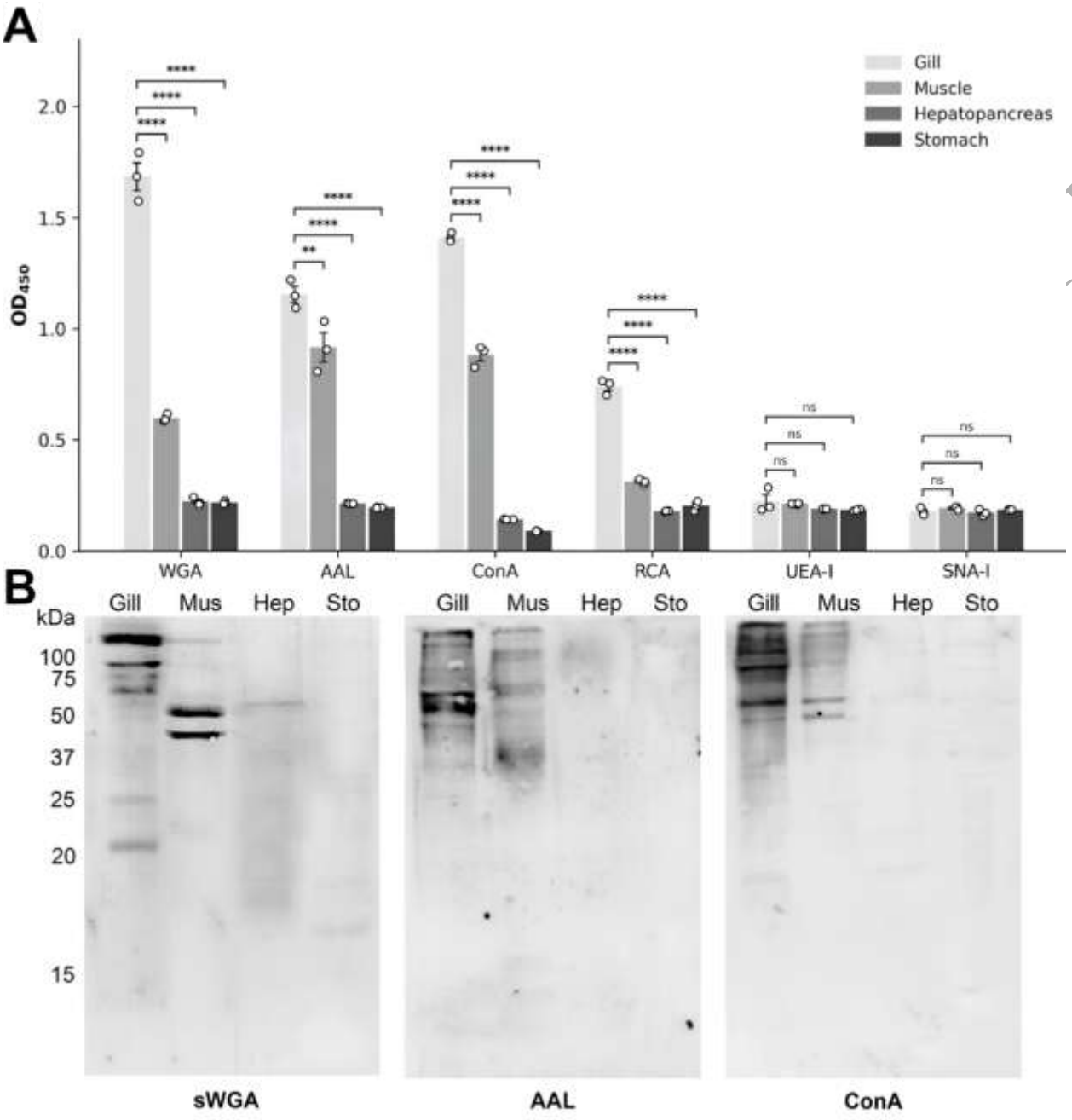



**Figure 3. Lectin–ELISA profiling indicates mannose- and fucose-rich glycans in shrimp tissue extracts**

(A) ELISA assays showing the binding of different lectins (WGA, AAL, ConA, RCA, UEA-I, and SNA-I) towards gill (Gil), muscle (Mus), hepatopancreas (Hep), and stomach (Sto) protein extracts. (B) Far-Western blot analysis of tissue extracts on PVDF membranes using biotin-



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conjugated WGA, ConA, and AAL. Data are presented as the mean values from three biologically independent samples. The bar graphs present the means of the reported data. Statistical comparisons were analyzed by one-way ANOVA followed by Dunnett's post-hoc test using gill as the reference control group (Ctrl) Significance levels are as follows: ****, $P \leq 0.0001$; *, $P \leq 0.05$; NS, $P > 0.05$.

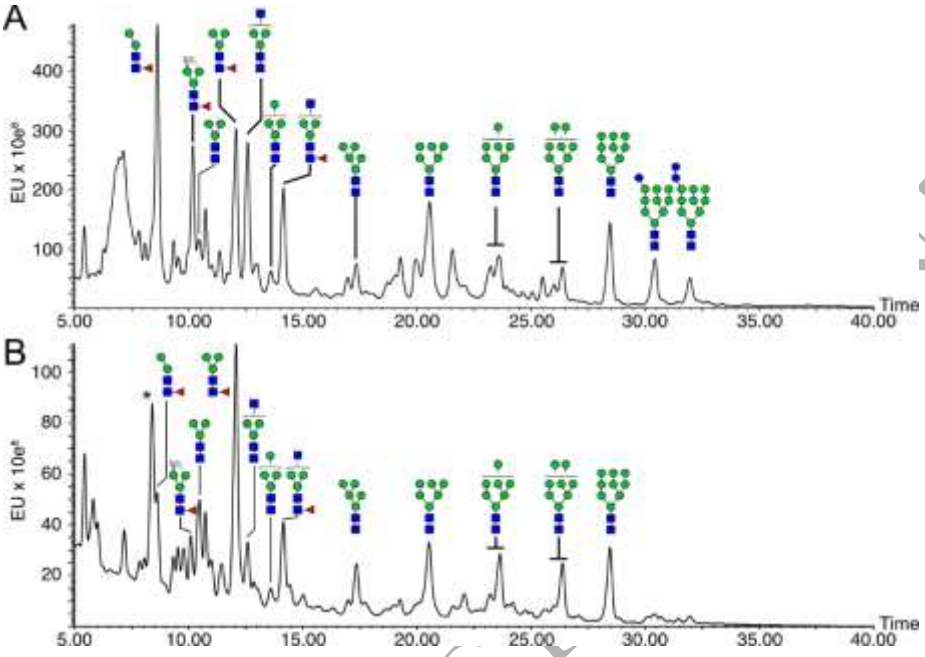



**Figure 4. *N*-glycomics profiles of muscle and gills analysed by LC-FL-MS of procainamide-derived *N*-glycans.** Fluorescence chromatograms of procainamide-labelled *N*-glycans released from (A) muscle and (B) gill tissues. The compositions of all signals were established by MS and the putative sequence by MS/MS fragmentation. Both tissues showed similar profiles of *N*-glycans characterized by the presence of pauci- and oligo-mannosylated glycans, albeit with differences in the ratios of individual glycans. * is a contaminant overlapping with H2H2F2signal.

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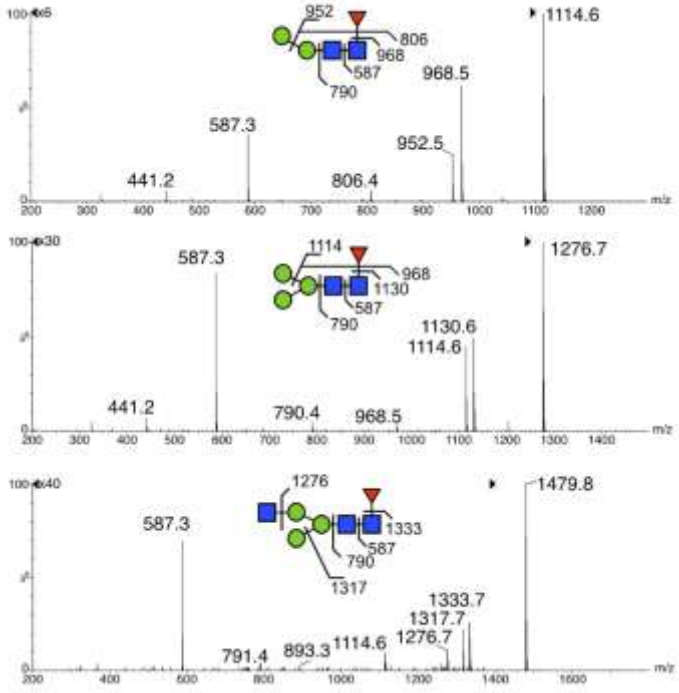



**Figure 5. Sequencing of core-fucosylated *N*-glycans observed in muscles.** LC-MS/MS analysis of procainanmide-labelled fucosylated *N*-glycans showed that they were all fucosylated on the first GlcNAc residue of the chitobiose core. MS/MS spectra were identical in muscles and gills.

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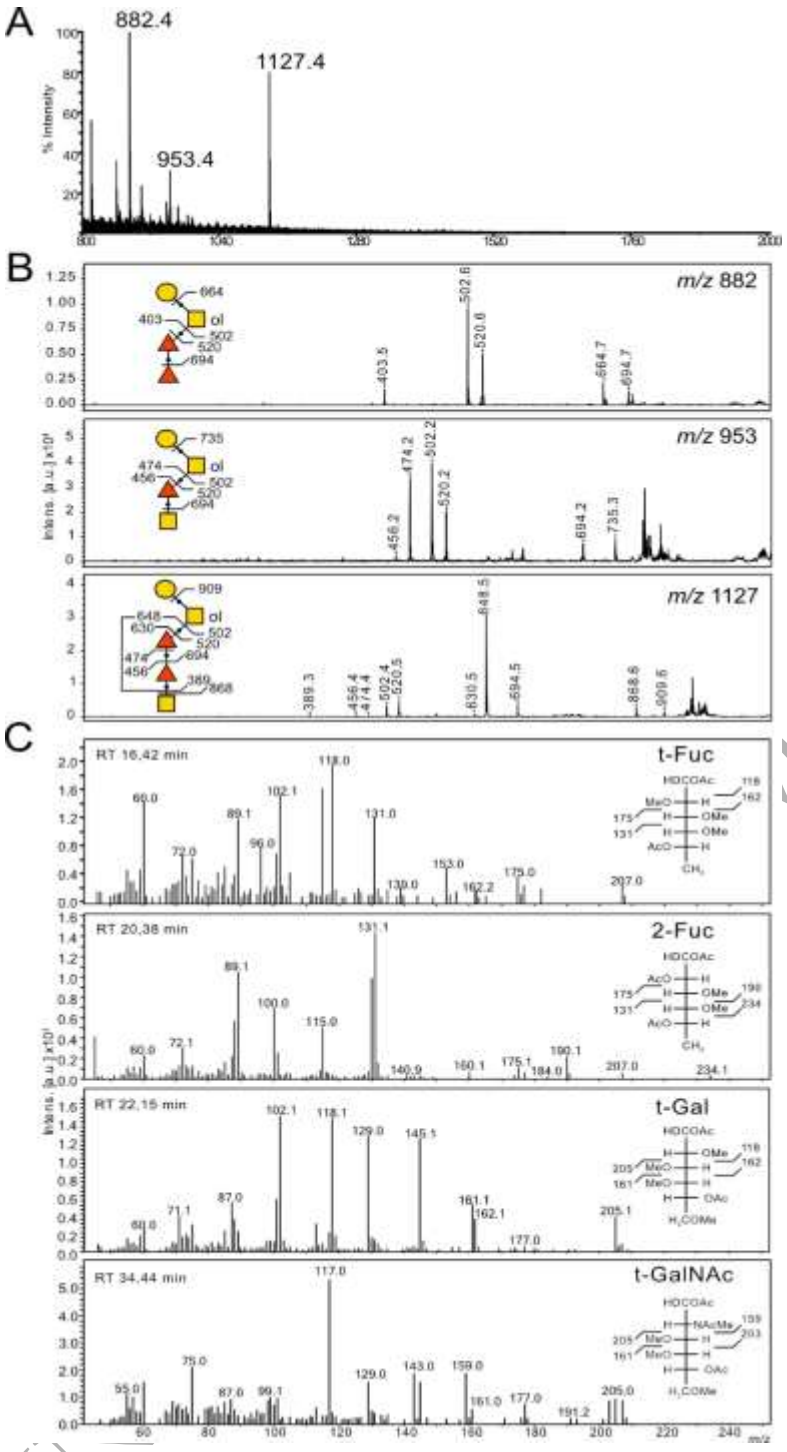



**Figure 6. *O*-glycomic profiles of gills analysed (A)** The MALDI-MS spectrum of permethylated *O*-glycan isolated from gills showed three major signals at *m/z* 882.3, 953.4 and 1127.5. **(B)** Sequence analysis by MALDI-MS/MS of signals as *m/z* 882.3 (top), 953.4 (middle) and 1127.5

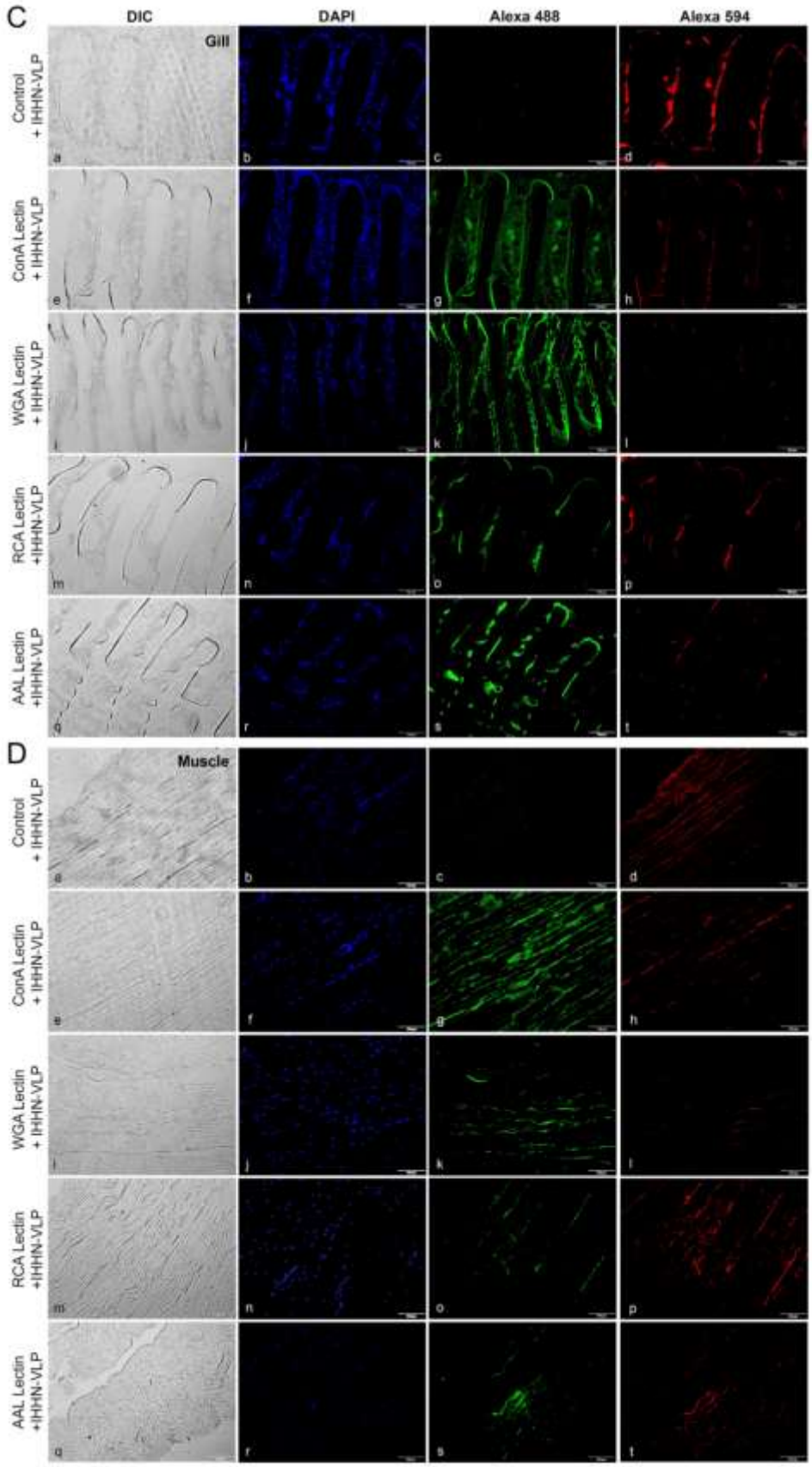
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(bottom) showed that *O*-glycans contain unusual motifs characterized by the presence of substituted internal fucose residues. **(C)** GC/MS analysis of PMAAs derived from *O*-glycans identified the internal fucose residues as 2-Fuc and established the presence of terminal fucose and *N*-acetylgalactosamine residues.

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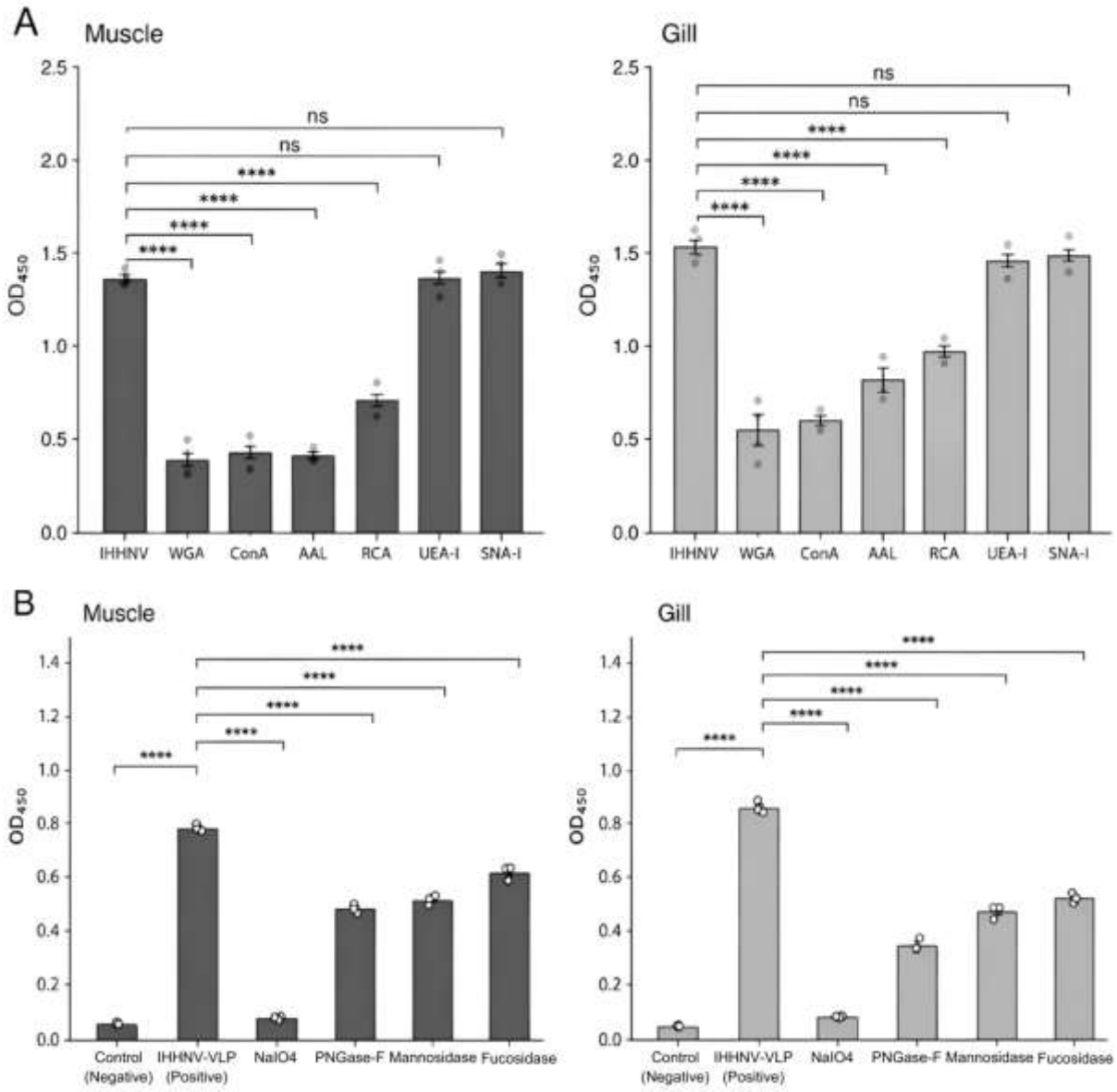



**Figure 7. IHHN-VLP binding and tissue lectin labeling** (A) Lectin-blocking ELISA of muscle (left panel) and gill tissues (right panel) showing that ConA, AAL, and WGA significantly inhibit IHHN-VLP binding, whereas UEA-I and SNA-I do not show significant effects. (B) Chemical or enzymatic disruption of glycans on muscles (left panel) and gills (right panel) lysates prior to VLP incubation. Periodate ($NaIO_4$) markedly reduced binding ($P < 0.0001$). PNGase-F, mannosidase, and fucosidase also significantly reduced binding ($P < 0.0001$), while

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fucosidase inhibition in muscle reduced less binding compared to enzyme (P<0.001). Data represent the mean ± standard deviation (SD) from three biologically independent samples. Statistical significance was assessed using one-way ANOVA with ****, $P \leq 0.0001$; ***, $P \leq 0.001$; NS, $P>0.05$. Immunofluorescence staining of gill (C) and muscle tissues (D) by IHHN-VLP (red) with or without pre-incubation with biotinylated ConA, WGA, RCA and AAL lectins (green).

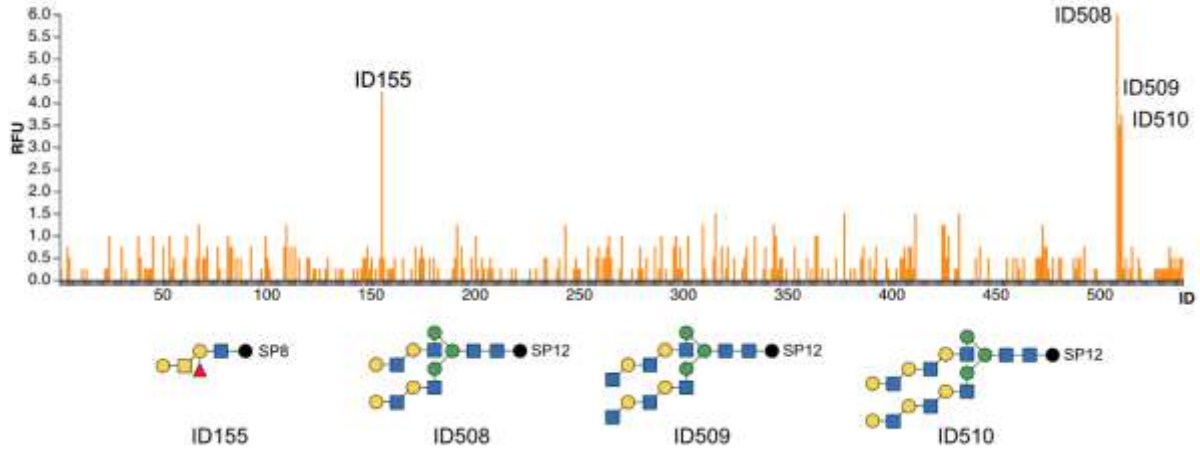



**Figure 8. Glycan specificity analysis of IHHNV-VLP on the CFG glycan microarray**. Microarray analysis of IHHNV VLP showed a very restricted specificity toward mammalian-like glycan motifs essentially concentrated on type-2 oligo-LacNAc substituted N-glycans (ID508-510). Full data including the Chart ID, sequences of the glycans, average RFU, StDev, %CV and numbers of spots are accessible in the Supplementary Table 1.

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## Tables

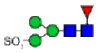
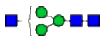
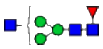
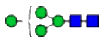
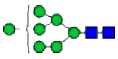
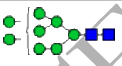
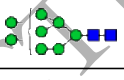
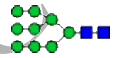
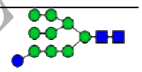
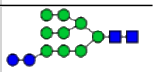
| RT | *m/z* | *Theo. m/z* | Composition | Structure |
|---|---|---|---|---|
| 8,63 | 1114,60 | 1114,5131 | N2H2F1 |  |
| 10,19 | 1356,56 | 1556,6757 | N2H3F1SO1 |  |
| 10,37 | 1130,60 | 1130,5080 | N2H3 |  |
| 12,07 | 1276,66 | 1276,5659 | N2H3F1 |  |
| 12,63 | 1333,68 | 1333,5874 | N3H3 |  |
| 14,15 | 1479,76 | 1479,6453 | N3H3F1 |  |
| 13,62 | 1292,65 | 1292,5608 | N2H4 |  |
| 17,35 | 1454,72 | 1454,6136 | N2H5 |  |
| 20,55 | 1616,76 | 1616,6664 | N2H6 |  |
| 23,63 | 1778,80 | 1778,7192 | N2H7 |  |
| 25,93 | 1940,88 | 1940,7720 | N2H8 |  |
| 26,38 | 1940,88 | 1940,7720 | N2H8 |  |
| 28,45 | 2102,98 | 2102,8248 | N2H9 |  |
| 30,43 | 2265,03 | 2265,0306 | N2H10 |  |
| 31,90 | 2427,15 | 2427,0834 | N2H11 |  |

**Table 1** Composition in *N*-glycans of gills and muscles established by LC/MS analysis of procainamide labelled *N*-glycans. RT, retention times in minutes; observed *m/z* mass over charge ratios are expressed for $[M+H]^+$ adducts; calculated theoretical *m/z* mass over charge ratios are expressed for $[M+H]^+$ adducts; compositions are expressed in number of HexNAc (N), Hex (H), Fucose (F) and sulfate (SO) residues; structures are designed in accordance to SFNG (Neelamegham et al. 2019)

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